

Exercise 3:

“Generate a detailed report on how you can use generative AI as a university student to learn subjects like probability and statistics, data base management systems and software engineering. Please write the entire report in English.”

DeepSeek (DeepThink R1)

Generative AI as a Learning Companion for University Students: A Report on Probability & Statistics, Database Management Systems, and Software Engineering

Executive Summary

Generative Artificial Intelligence (GenAI) has rapidly transformed the higher education landscape, with nearly half of all studies on AI in education targeting STEM subjects. For university students navigating demanding courses in probability and statistics, database management systems (DBMS), and software engineering, GenAI tools offer unprecedented opportunities for personalized, interactive, and accelerated learning. This report provides a comprehensive analysis of how students can strategically leverage generative AI across these three domains, detailing specific use cases, practical prompts, recommended tools, and critical considerations for responsible use.

1. Probability and Statistics

1.1 Overview of Challenges

Probability and statistics present unique learning hurdles for university students. Many students struggle with visualizing how probability density functions (PDFs) change as parameters shift, calculating critical values without cumbersome calculators or dense textbook appendices, and translating statistical values into a clear understanding of rejection regions for hypothesis testing. These abstract concepts often feel disconnected from practical application, making retention and comprehension difficult.

1.2 Key Applications of Generative AI

Concept Explanation and Definition. Generative AI excels at providing clear, accessible explanations of statistical concepts. Students can begin by asking AI to define key statistical terms in plain language and then compare those definitions with textbook or lecture materials. This comparative approach deepens understanding by exposing students to multiple explanatory frameworks.

Step-by-Step Problem Solving. AI can walk students through the logic of hypothesis testing, explaining when to use which test (t-test vs. ANOVA vs. chi-square) and guiding them through the reasoning process. Students can present a problem and ask the AI to solve it step-by-step, comparing the AI's approach with their own manual calculations.

Data Generation and Simulation. One of the most powerful applications is using AI to generate random datasets for practice. For instance, students can prompt: *"Generate 100 random values from a normal distribution with mean 50 and standard deviation 10, then calculate the probability of values above 60"*. This allows students to practice with fresh data repeatedly, reinforcing their skills without exhausting textbook examples.

Interactive Visualization. Generative AI can be used to create interactive web applications that bring statistical distributions to life. Using what some educators call "vibe coding"—describing desired functionality in natural language and letting AI generate the code—students can build tools that visualize distributions like the t-distribution, chi-squared distribution, and F-ratio distribution with adjustable parameters and visual highlighting of rejection regions.

Practice Problem Generation. AI can serve as a personalized problem generator. By prompting the AI to act as a patient instructor and provide concrete scenarios involving probabilities, students receive unlimited practice problems tailored to their current level of understanding. Templates exist for generating assessments across data classification, descriptive statistics, probability, and inferential statistics.

1.3 Research Evidence

A study evaluating ChatGPT's effectiveness in solving probability problems from computer engineering exams found that ChatGPT outperformed the average student in writing, organization, and logical reasoning. While the model showed some difficulties with basic numerical operations, these limitations could be overcome by requesting solutions via programming scripts. The study concluded that the model demonstrates high potential as a learning assistant due to its ability to provide high-quality explanations and illustrate solutions in multiple programming languages.

1.4 Recommended Tools

- MathBot: An agentic AI tutor developed for Babson College, designed specifically to support students with complex topics in probability and statistics through personalized, academically rigorous tutoring.
- StatChat: A secure RAG-based AI tutor for engineering statistics courses that assists with homework, document retrieval, and programming assistance.
- General-purpose LLMs: ChatGPT, Claude, and Gemini can all serve effectively for probability and statistics learning when prompted appropriately.

2. Database Management Systems (DBMS)

2.1 Overview of Challenges

Database management is a critical field within information systems, requiring students to master a wide array of skills—from conceptual design to SQL programming and real-world problem-solving. Traditional teaching methods often present database concepts in isolation, hindering students' ability to grasp the holistic nature of database design, development, and maintenance. Additionally, current pedagogical approaches tend to emphasize technical syntax over critical thinking and provide limited feedback, especially in larger classes.

2.2 Key Applications of Generative AI

SQL Query Generation and Explanation. Students can describe what they want to accomplish in natural language, and AI generates the corresponding SQL query while explaining each component. This is particularly valuable for beginners who may understand the logic of what they want to retrieve but struggle with syntax. Conversely, students can paste a complex SQL query and ask the AI to explain it step-by-step.

Database Design and Normalization. AI can guide students through the normalization process, helping them apply principles of First, Second, and Third Normal Form (1NF, 2NF, 3NF) to eliminate redundancy and optimize relationships. Students can prompt: *"Review the following PostgreSQL schema for normalization issues. Identify any violations of 1NF, 2NF, or 3NF, explain why each is a problem, and suggest a corrected structure"*. This transforms abstract normalization theory into concrete, actionable practice.

Immediate Feedback and Iterative Learning. One of the most significant advantages of AI in DBMS education is the provision of immediate, personalized feedback. Students can attempt to write queries or design schemas, receive AI feedback, refine their work, and resubmit—all without waiting for office hours or graded assignments. This accelerates the learning cycle and allows students to explore more advanced database operations at their own pace.

Real-World Scenario Practice. AI can generate dynamic business requirements and ask students to design relational databases, generate SQL queries, and maintain databases in response to these evolving needs. This bridges the gap between textbook learning and practical application.

Practice Database Generation. Students can ask AI to create sample databases with realistic schemas and data, providing them with a sandbox environment for practicing queries, joins, subqueries, and transactions.

2.3 Research Evidence

A teaching case study integrating ChatGPT into a database management capstone project found significant performance improvements in graduate cohorts, with undergraduate cohorts showing a positive upward trend. Students using ChatGPT were transformed from passive recipients of information into critical thinkers who could apply their skills to real-world problems. A separate study with 37 computer science students who used ChatGPT to learn database administration found that high-performing students were the ones using ChatGPT most extensively. This suggests that while AI can benefit all students, those who are already engaged and motivated derive the greatest benefit.

2.4 Recommended Tools

- **DBMS Learning Assistant:** An AI-powered tool that helps students learn DBMS concepts interactively with practical SQL query execution, featuring natural language to SQL conversion, step-by-step explanations, and real database practice.
- **QueryMind:** An AI SQL learning assistant that allows students to ask database questions in plain English and get help writing and optimizing queries.
- **ReviseMate:** An AI-powered exam revision buddy that provides quizzes and instant feedback on DBMS topics.
- **General-purpose LLMs:** ChatGPT, Claude, and Gemini can all assist with SQL generation, schema design, and normalization explanation.

3. Software Engineering

3.1 Overview of Challenges

Software engineering education requires students to master not only coding but also requirements gathering, system design, project planning, testing, debugging, and team collaboration. The field demands both technical proficiency and higher-order thinking skills such as analysis, integration, and problem-solving. Traditional education often focuses heavily on syntax and low-level implementation, leaving students ill-prepared for the complexities of real-world software development.

3.2 Key Applications of Generative AI

Code Implementation and Debugging. Research shows that GenAI is highly effective as a "companion" for automating low-level tasks such as code implementation, debugging, and test scenario specification. This allows students to focus on higher-level design and decision-making rather than getting bogged down in syntax errors and boilerplate code. Students can present buggy code to AI and ask for debugging assistance, learning not just the fix but the reasoning behind it.

Requirements Gathering and Documentation. AI can assist in conducting thorough requirements gathering for software projects. Students can leverage AI to identify knowledge gaps, create comprehensive Product Requirements Documents, and uncover crucial details through iterative questioning. This is particularly valuable for project-based courses where students must define the scope and functionality of their software before beginning implementation.

Project Planning and Task Breakdown. LLMs can transform high-level problem statements into structured, sprint-ready software development plans. Students can describe their project goals and receive a breakdown of epics, stories, tasks, and estimated timelines—developing project management skills alongside technical ones.

Test Scenario Generation. AI can automatically generate test cases and test scenarios based on functional specifications. This helps students develop robust testing practices and understand the importance of quality assurance in the software lifecycle.

Code Review and Optimization. Students can submit their code for AI review, receiving suggestions for optimization, style improvements, and potential edge cases they may have overlooked. This mimics the experience of peer or professional code review, building valuable skills in code quality assessment.

Design and Architecture Guidance. AI can help students reason about system architecture, suggesting appropriate design patterns, data structures, and architectural approaches based on project requirements.

3.3 Research Evidence

A two-year teaching experience integrating GenAI into a coordinated software development project for fourth-year Computer Science students found that GenAI allowed students to focus on higher-level design and decision-making. Initial findings revealed student frustration when GenAI struggled with advanced project contexts, often due to a lack of prompt expertise. However, corrective training actions—including seminars on prompt engineering and critical analysis—significantly improved outcomes. Usage patterns were asymmetric, with general-purpose tools dominating early stages and specialized tools gaining relevance as project complexity increased. This experience confirmed the value of GenAI in promoting critical and responsible use within software engineering education.

3.4 Recommended Tools

- GitHub Copilot: The most reliable autocomplete tool for coding, integrated directly into development environments.
- Cursor: An AI-powered IDE that provides intelligent code completion and assistance.
- Claude Code: A terminal-based, agentic tool effective at working through multi-step problems.
- Gemini Code Assist: Offers a generous free tier, making it particularly accessible for students.
- General-purpose LLMs: ChatGPT remains the most widely used GenAI tool among students, with 74% of respondents in one study reporting its use.

4. Cross-Cutting Strategies for Effective GenAI Use

4.1 The Importance of Prompt Engineering

Across all three domains, the quality of AI output depends critically on the quality of the prompt. Students who write targeted, precise prompts achieve stronger outcomes. Vague requests such as "Explain the normal distribution"

produce far less useful results than specific prompts like *"Generate 100 random values from a normal distribution with mean 50 and standard deviation 10, then calculate the probability of values above 60"*. As one student reflection noted, "a good prompt is the key to good results".

4.2 Critical Thinking and Verification

Generative AI models have a propensity to provide inaccurate or made-up information—a phenomenon known as "hallucination". In educational contexts, hallucinations risk misleading students, compromising academic integrity, and eroding trust in AI-assisted learning. Students must therefore maintain critical thinking and verify AI outputs against textbooks, lecture notes, and other reliable sources.

Some educators have turned this challenge into a pedagogical opportunity, having students detect hallucinations and biases in AI outputs as a vehicle for developing critical thinking skills. Rather than passively accepting AI-generated content, students should question, verify, and compare.

4.3 Responsible Use and Academic Integrity

The integration of GenAI in education requires a shift in teaching methodologies—from prohibition and penalization to encouraging virtuous use while limiting risks. Students should understand that GenAI is a learning companion, not a replacement for learning. Key principles include:

- Using AI to enhance understanding, not to avoid learning
- Always verifying AI-generated content
- Disclosing AI use where required by course policies
- Developing prompt engineering skills alongside domain knowledge

4.4 Recommended Learning Workflow

A structured approach to using GenAI for learning might include:

1. Initial Exploration: Use AI to get overviews and definitions of new concepts
2. Guided Practice: Generate practice problems and work through them with AI assistance

3. Self-Assessment: Compare your solutions with AI-generated solutions
 4. Verification: Cross-check AI outputs with authoritative sources
 5. Reflection: Write reflections on what you learned from AI versus traditional instruction
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5. Conclusion

Generative AI represents a transformative tool for university students studying probability and statistics, database management systems, and software engineering. In probability and statistics, AI enables interactive visualization, data generation, and personalized problem-solving. In DBMS, AI facilitates SQL learning, database design, and normalization practice with immediate feedback. In software engineering, AI automates low-level coding and debugging tasks, allowing students to focus on higher-order design and project management skills.

However, the effectiveness of GenAI depends on intentional, responsible use. Students must develop prompt engineering skills, maintain critical thinking, verify AI outputs, and use AI as a supplement to—not a replacement for—traditional learning methods. When used thoughtfully, generative AI can accelerate learning, deepen understanding, and prepare students for the AI-augmented workplaces of the future.

The evidence is clear: GenAI is not a threat to education but an opportunity—one that, when harnessed correctly, can transform students from passive recipients of information into active, critical, and creative learners equipped for the challenges of the 21st century.

Qwen (Thinking)

Generate a detailed report on how you can use generative AI as a university student to learn subjects like probability and statistics, data base management systems and software engineering.

💡 Thinking completed >

Comprehensive Report: Leveraging Generative AI for Academic Excellence in Technical University Subjects

Prepared for: University Students in Computer Science, Data Science, and Information Systems

Subject Focus: Probability & Statistics, Database Management Systems (DBMS), Software Engineering (SE)

1. Executive Summary

Generative AI (GenAI) has transitioned from a novelty to a fundamental tool in higher education. For highly technical and conceptual subjects like Probability & Statistics, DBMS, and Software Engineering, GenAI acts as a personalized tutor, a coding co-pilot, and a system design consultant. This report outlines actionable strategies, subject-specific applications, and ethical best practices for university students to maximize their learning outcomes using GenAI without compromising academic integrity or foundational understanding.

2. Core Methodologies for AI-Assisted Learning

Before diving into specific subjects, students must adopt the right pedagogical approach when interacting with GenAI.



+ Ask Qwen

Thinking ▾



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Grok (DeepSearch)

Report Overview

Title: *Using Generative AI as a University Student*

Subtitle: A Practical Guide to Learning Probability & Statistics, Database Management Systems, and Software Engineering

Structure

1. **Introduction** — Why generative AI matters for university learning, especially for CS/STEM students (drawing on large-scale usage studies).
2. **Core Principles for Effective & Ethical AI Use**
 - Treat AI as a collaborator, not an oracle
 - Understand first, then accelerate
 - Verify everything
 - Iterate with multi-turn conversations
 - Disclose according to policy
 - Protect privacy
3. **Probability and Statistics**
 - Multiple explanations of concepts (formal + intuitive + numerical)
 - Guided problem-solving (hints instead of full answers)
 - Generating practice problems and quizzes
 - Simulation, visualization, and code (Python/R)
 - Support for advanced topics (Bayesian methods, etc.)
 - Concrete example prompts included

Ask anything



Fast ▾



Reflection

Doing this exercise showed me that AI can be much more than just a quick answer generator. As an IT student, I really appreciated how these models broke down Software Engineering concepts and explained Database queries step-by-step. Even for subjects like Probability and Statistics, seeing the AI's reasoning process made the math much easier to follow. It basically felt like having a study buddy who explains the actual logic instead of just handing over the final result.

Best

In my opinion, **DeepSeek (DeepThink R1)** produced the best results. Its advanced reasoning process provided a highly structured and detailed report, making it incredibly easy to understand exactly how AI can be applied to complex mathematical and IT subjects.